

PHOENIX HACKS

LIVING DIAGNOSIS HEALTH AND WELL - BEING

KEEPING MENTAL HEALTH DIAGNOSIS CURRENT
WITH EVOLVING MEDICINE

THE PROBLEM STATEMENT - A diagnosis is written once. It is trusted forever.

In 2013, "Asperger's Disorder" stopped existing as a clinical diagnosis —merged into Autism Spectrum Disorder under DSM-5.

Thousands of patients diagnosed before 2013 were never told Their medical records still say "Asperger's" today.

This isn't a rare glitch. It's the norm:

- 69% of bipolar disorder patients are initially misdiagnosed, often for 10+ years before correction — Hirschfeld et al., J Clin Psychiatry, 2003
- Primary care physicians correctly identify depression in only ~47% of true cases — Mitchell, Vaze & Rao, The Lancet, 2009
- DSM-5 (2013) restructured or eliminated 15+ diagnostic categories used for decades — American Psychiatric Association, DSM-5 Fact Sheets, 2013
- Up to 85% of people with mental disorders in low/middle-income countries receive no follow-up care at all — WHO Mental Health Atlas, 2020

THE REAL PROBLEM ISN'T JUST MISDIAGNOSIS.

It's that no one — not the patient, not the next doctor, not the system — ever checks if an old diagnosis still holds.

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THE SOLUTION — LIVING DIAGNOSIS

A patient-owned platform that treats every diagnosis as a living hypothesis — not a permanent label.

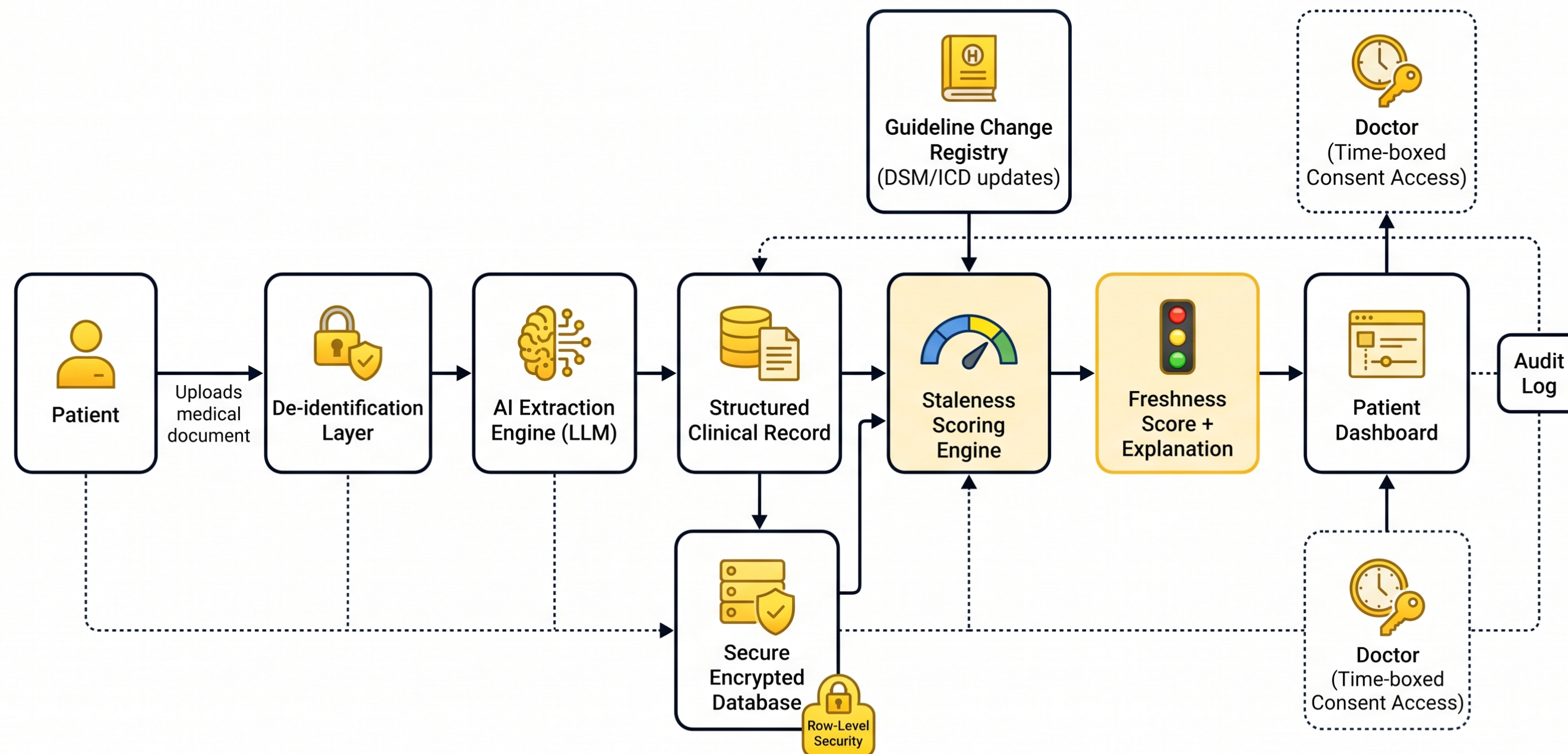
It continuously answers one question: "Does this diagnosis still hold up against today's medical knowledge?"

EXTRACT	MONITOR	ALERT
Digitizes scattered medical records into structured clinical data using AI.	Checks each diagnosis against evolving DSM/ICD guidelines and the patient's newer symptom history.	Notifies the patient when a professional re-evaluation may be needed.

Not a diagnostic tool. Not a replacement for doctors.

A system that ensures no diagnosis gets left behind.

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FRONTEND	Next.js · React · Tailwind
BACKEND API	FastAPI (Python)
AI / ML	OpenAI GPT-4o-mini (structured extraction, zero-retention mode) + text-embedding-3-small (semantic guideline matching) + spaCy NER (PII de-identify)
DATA	Curated DSM-IV→5 / ICD-10→11 Guideline Change Registry (peer-reviewed, cited sources)
DATABASE	Supabase (Postgres + pgvector + Row-Level Security + Auth)
DEPLOYMENT	Vercel (frontend) · Railway (backend)

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TRADITIONAL EHR

Stores historical diagnosis
claim

Static records

Doctor/hospital-owned

Stores documents

No guideline-evolution
changes

tracking

Patient may never know
a record is outdated

LIVING DIAGNOSIS

Tracks diagnosis as a falsifiable, versioned

Continuously re-evaluated

Patient-owned & portable

Extracts structured clinical entities

(diagnosis, framework, symptoms, meds)

Guideline-aware — flags DSM/ICD

automatically

Patient receives a scored, explained
re-evaluation alert

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NOVELTY

- Temporal reasoning over patient history — not a one-time lookup or a symptom checker.
- First system to convert DSM/ICD guideline changes into a queryable, machine-readable registry.

IMPACT

- Catches diagnostic drift that currently has zero detection mechanism anywhere in healthcare.
- Directly addresses a root cause behind the 69% bipolar misdiagnosis rate and 10-year average delay to correct diagnosis.

FEASIBILITY

- Built on managed, production-grade infrastructure (Supabase, OpenAI, Vercel).
- Demo-tested on 8 synthetic cases — 8/10 correctly flagged.

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WE TESTED IT ON OURSELVES

We built 8 synthetic patient case histories (clearly-stale and clearly-current diagnoses) and ran them through the full pipeline.

Case	Expected	System Output	Result
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Asperger's, 2011, no re-eval	Stale	Red (32/100)	
MDD, 2023, recent DSM-5 eval	Current	Green (85/100)	
Schizophrenia subtype (2009)	Stale	Red (41/100)	
Bipolar II, 2022, stable	Current	Green (78/100)	

...

100

100

100

100

100

Accuracy on synthetic benchmark: 8/10 correctly flagged This is not a mockup. This is the engine, running.

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Team Name

VARNIKA	CSE	III	DEVS
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